REPORT LAB 4

THERMOCOUPLE

Class: CC10 / Group: 1		Lecturer's comment
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I. Aims/Purposes

- To get used to thermocouple (chromel-alumel) MC-897 electronic millivoltmeter to determine the thermocouple constant.

II. Apparatus, Methods, and Procedure

a) Apparatus

- Thermocouple (chromel-alumel);
- Thermocouple holder;
- MC-897 electronic millivoltmeter;
- Two 250ml experiment glasses;
- Digital thermometer (-50 ÷ 150°C);
- Wine thermometer (0 ÷ 100°C);
- Boiler (220V - 300W);
- Digital multimeter 3,1/2 Digits
- Set of wires.

b) Methods

- Using indirect method. Measuring the temperature with the thermocouple then plot the graph and calculate the time constant using the formula.
- There are two methods to determine the thermocouple constant.

c) Procedure

- Step 1:
  + Set the range for millivoltmeter mV at position 15.
  + Set the digital multimeter at the measurement range of 200mVDC.
  + Use a piece of wire to short circuit the positive and negative probes.
  + Press button K on the front and millivoltmeter mV starts. Turn sensitivity knob R counterclockwise all the way left.
  + Turn knob "0" until the reading is 5mV on the electronic millivoltmeter, then read the value of $U_{k\text{đ}}$ on the digital multimeter to derive the amplification coefficient k = $U_{k\text{đ}}$ /5 mV. Afterward, repeat zeroing for milivoltmeter mV, disconnect the wire from the positive and negative probes and connect it to the thermocouple.

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- Step 2:
  + Place the bulbs of two thermometers in the middle of boiler and experimental glass at the same level as the connections of thermocouples .
  + Wait 4-5 minutes before monitoring and recording the temperature T_1 on thermometer in the boiler and temperature T_2 on thermometer in the glass.
  + If T_1 equals T_2, the milivoltmeter's mV pointer must point to 0.
- Step 3:
  + Connect the boiler to the 220V mains. Wait until the water in the boiler starts to boil before unplugging the boiler and allowing the water to cool (avoid a quick cool down).
  + Record the reading of $U_{kđi}$ on the digital multimeter for each value of T_1 to compute the thermal electromotive force E(n) and write it down. Repeat this procedure until the temperature T_1 falls below 40°C.
  + *Note:* When the range selector "RANGE" of millivoltmeter mV is set at X10 and sensitivity knob R is turned all the way left, the maximum limit on millivoltmeter mV is 15mV.
- Step 4:

Read and record the following data in Table 1:

  + The level of accuracy δ_v and the maximum value U_m on the measurement range of electronic.
  + milivoltmeter mV.
  + The level of accuracy δ_v and the maximum value U_m on the measurement range of digital.
  + milivoltmeter mV.
  + Accuracy ΔT of thermometers 3 and 4.

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III. Equations

$$k = \frac{U_{digital}}{U_{analog}}$$

$$\frac{\Delta k}{k} = \frac{\Delta U_{digital}}{U_{digital}} + \frac{\Delta U_{analog}}{U_{analog}}$$

$$\Delta U_{analog} = \delta_v \cdot U_{analog\ range}\ (mV)$$

$$\Delta U_{digital} = \delta_v \cdot U_{digital} + n\alpha\ (mV)$$

$$E_i = \frac{U_{kđ\ i}}{k}$$

$$\frac{\Delta E_i}{E_i} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ i}}{U_{kđ\ i}} = \varepsilon_i$$

$$\Delta E_i = \varepsilon_i E_i\ (mV)$$

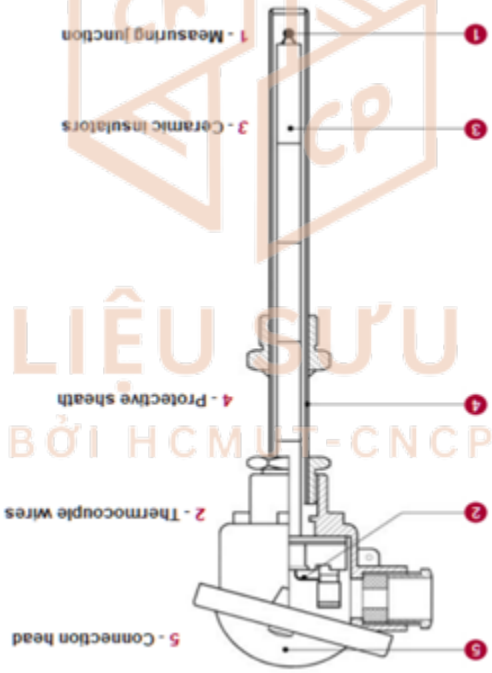
$$\Delta(T_1 - T_2) = \Delta T_1 + \Delta T_2\ (°C)$$

IV. Question

1. Present the structure of a thermocouple and the cause of thermal electromotive force within a thermocouple.

Structure of a thermocouple:

- Measuring the junction: this is the most important part of the thermocouple. This element will be made up of two essence rods with differing material structures that will be welded together at one end.
- Thermocouple wires: the wire that connects the temperature sensing component to the controller.
- Ceramic insulators: this is the ceramic insulator used to keep the thermocouple line isolated along the entire length of the inquiry.
- Protective sheath: This is the thermocouple's exterior protective shell. For temperatures of 1200 degrees Celsius or less, this shell is usually composed of stainless steel. This shell is constructed of porcelain for thermocouples with a scale higher than 1200 degrees Celsius.
- Connection head: this part will contain the connection line of the thermocouple. When we use thermocouple transformers, 4-20mA will be input then.



The cause of thermal electromotive force within a thermocouple.

As seen in the illustration below, A and B are two different cables that are joined to form a junction in the circuit of a thermocouple. The Peltier electromotive force in the circuit is a function of the temperatures of two junctions, and it's created when the temperatures of the two junctions differ.

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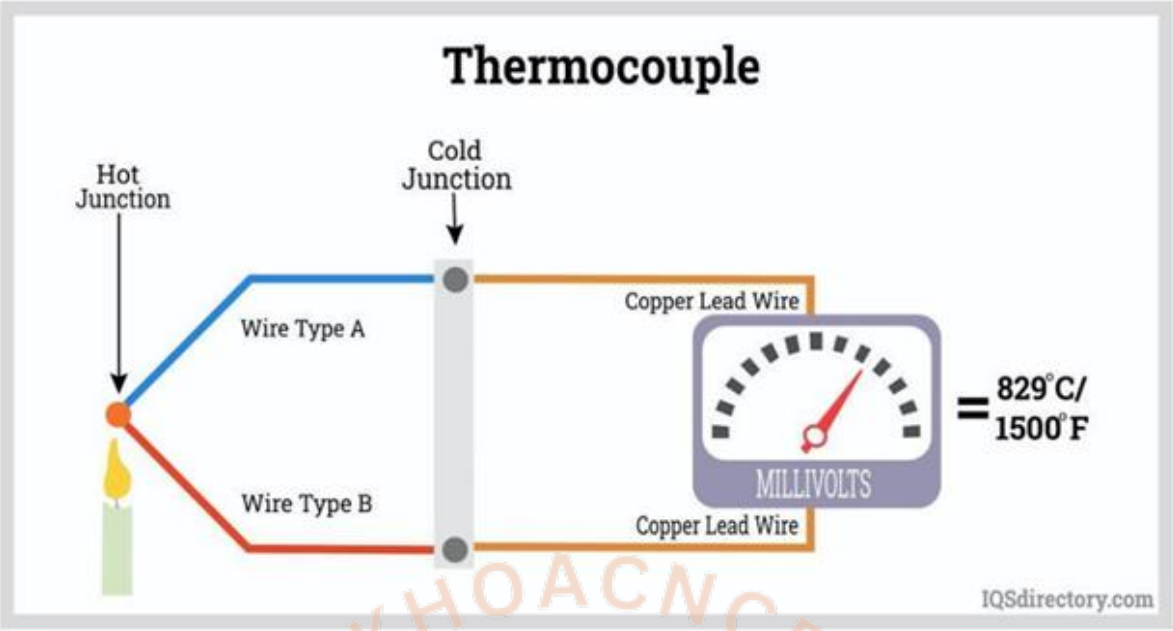

Electrons transport heat and electricity. When one end of a copper wire is heated, electrons move down the wire to the cooler end, resulting in a temperature gradient. The heat has been transformed into energy. The same idea that V olta and Seebeck discovered also applies to thermocouples.

When the temperature of the thermocouple's connections is the same, the junctions create an equal and opposite EMF, and the current flow is zero. If the junctions have differing temperatures, the EMF will not be zero, and current will flow through the circuit in the same manner as heat does through copper wire.

The EMF in the thermocouple circuit is very small, measured in millivolts, therefore detecting the generated electromotive force necessitates a highly sensitive instrument. Normally, galvanometers and voltage balancing potentiometers are employed, with the potentiometer being the most common.

Seebeck effect: The Seebeck effect occurs when two distinct or unlike essences are linked at two junctions and an electromotive force is generated at the two junctions, which varies depending on the essence.

Peltier effect: An electromotive force is generated in a circuit when two dissimilar metals are joined to form two junctions due to the different temperatures of the two junctions of the circuit.

2. What factors does the magnitude of thermal electromotive force depend on? State clearly the relation between thermal electromotive force *E* and the temperature difference ($T_1 - T_2$) between the two junctions of a thermocouple.

Factors that influence the amount of thermal electromotive force include:

- The thermocouple is a part of the nature of metals.
- The difference in temperature between the hot and cold connectors.
- The relationship between the temperature differential and the thermal electromotive force E.
- The link between temperature and electromotive force is monotonic and non-linear, which means that when the battery's emf grows, temperature rises as well. It also does

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not mean that the emf is always rising. Every year, it does not rise at the same rate. The aforementioned link between the battery's emf and the Gibbs free energy term may be determined using the Gibbs Helmholtz equation.

$$E = \frac{\Delta G}{nf} = \frac{\Delta S}{dT \times nf}$$

- The temperature affects the amount of current induced. A heated solenoid, for example, will push/pull less electrons than a cool solenoid. In both circumstances, the emf remains constant, as does the labor being done.

3. Why is sampling needed prior to the use of a thermocouple to measure temperature?

- The reference junction temperature has to be known to get an accurate absolute temperature reading because the thermocouple is a discriminational device rather than an absolute temperature dimension device.
- Reference junction compensation is the term for this procedure (cold junction compensation.)

4. Why is the cooling-down process instead of the heating-up process of the hot junction selected to sample a thermocouple? Sampling a thermocouple is a process to investigate the dependence of thermal electromotive force on the temperature difference between the two junctions as denoted in equation $\boldsymbol{E = C(T_1 - T_2)}$.

- The thermocouple requests more heat because the heater is surrounded by a coking, or insulating, layer of oil and the heat is not transmitting efficiently into the process. This causes the heater to run hotter, raising the element temperature and adding to the coking layer, which inhibits heat from entering the process and sets off a vicious cycle. A system failure is unavoidable without a mechanism to recognize this condition.
- The millivoltmeter's resistance RV is significantly greater than the thermocouple's internal resistance r, we can apply Ohm's law to a closed circuit and get:

$$E = I(R_v + r) \approx IR_v = U_v$$

where I is the current intensity and Uv is the millivoltmeter value in millivolts (mV).

- Thus, we measure the equivalent value of thermal electromotive force E on millivoltmeter mV for each value of temperature differential (T1 - T2) between the hot junction (T1) and the cold junction (T2) of the thermocouple. As a result, we may build the thermocouple sample graph and calculate the value of constant C.

5. Present how to determine thermocouple constant C using a graph to demonstrate the dependence of thermal electromotive force on the temperature difference between the two junctions of the thermocouple as denoted in equation $\boldsymbol{E = C(T_1 - T_2)}$?

- Prior to the appetite of a thermocouple, we've to sample the thermocouple; that is, we've to probe the temperature range (calculating on the thermocouple type) where thermal electromotive force is linearly contingent on the temperature change between its two corners. As a result, the graph symbolizes the function of is a member (Figure 2).
- Constant C is counted as in Equation 2:

$$C = \frac{En\text{đ}}{T_1 - T_2} = tg\alpha$$

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6. State clearly advantages of a thermocouple overa liquid thermometer(wine ormercury) in temperature measurement.

- Unparalleled accuracy.
- Low-cost and extremely dependable.
- Temperature range: Thermocouples are capable of handling temperatures ranging from cryogenics to jet engine exhaust. Depending on the metal wires used, a thermocouple may monitor temperatures ranging from –200°C to +2500°C.
- Sturdy: Thermocouples are shock and vibration resistant, making them excellent for use in hazardous environments.
- Because thermocouples are small and have a low thermal capacity, they respond fast to temperature changes, especially if the sensing junction is exposed. They can react to rapidly changing temperatures in a few hundred milliseconds.
- No self-heating: Thermocouples are naturally safe and do not self-heat because they do not require excitation power.

V. Experimental data

Table 1	- Milivolt-meter **mV**: δ_v = 1.5%; U_m = 15.0 *(mV)* - Digital milivolt-meter **mV**: δ_v = 0.5%; $n = 3$, $\alpha = 1$ - Alcohol thermometer: ΔT = 1 (°C) - Digital thermometer: ΔT = 0.3 (°C)					
Measurement/ T_2 = 28 (°C)	T_1 (°C)	$T_1 - T_2$	$\Delta(T_1 - T_2)$	$U_{kđ}$ (*mV*)	$E_{nđ}$ (*mV*)	$\Delta E_{nđ}$
1	95	67	1.3	152.6	5.09	0.300
2	90	62	1.3	132.4	4.41	0.260
3	85	57	1.3	119.3	3.98	0.239
4	80	52	1.3	105.2	3.51	0.211
5	75	47	1.3	96.2	3.21	0.193
6	70	42	1.3	84.7	2.82	0.170
7	65	37	1.3	74.9	2.50	0.153
8	60	32	1.3	67.7	2.26	0.138
9	55	27	1.3	58.4	1.95	0.121
10	50	22	1.3	48.1	1.60	0.101
11	45	17	1.3	39.8	1.33	0.086

VI. Calculating values and uncertainties

Amplifier coefficient:

$$k = \frac{U_{digital}}{U_{analog}} = \frac{150}{50} = 30$$

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$\Delta U_{digital} = \delta_v \cdot U_{digital} + n\alpha = 0.5\% \times 150 + 3 \times 0.1 = 1.05\ (mV)$

$\Delta U_{analog} = \delta_v \cdot U_{analog\ range} = 1.5\% \times 15 = 0.225\ (mV)$

$$\frac{\Delta k}{k} = \frac{\Delta U_{digital}}{U_{digital}} + \frac{\Delta U_{analog}}{U_{analog}} = \frac{1.05}{150} + \frac{0.225}{5} = 0.052$$

$=> \ \Delta k = 0.052 \times k = 0.052 \times 30 = 1.56$

$$\frac{\Delta E_i}{E_i} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ i}}{U_{kđ\ i}} = \varepsilon_i$$

$$\frac{\Delta E_1}{E_1} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 1}}{U_{kđ\ 1}} = 0.052 + \frac{0.5\% \times 152.6 + 3 \times 0.1}{152.6} = 0.059$$

$$\frac{\Delta E_2}{E_2} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 2}}{U_{kđ\ 2}} = 0.052 + \frac{0.5\% \times 132.4 + 3 \times 0.1}{132.4} = 0.059$$

$$\frac{\Delta E_3}{E_3} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 3}}{U_{kđ\ 3}} = 0.052 + \frac{0.5\% \times 119.3 + 3 \times 0.1}{119.3} = 0.060$$

$$\frac{\Delta E_4}{E_4} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 4}}{U_{kđ\ 4}} = 0.052 + \frac{0.5\% \times 105.2 + 3 \times 0.1}{105.2} = 0.060$$

$$\frac{\Delta E_5}{E_5} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 6}}{U_{kđ\ 6}} = 0.052 + \frac{0.5\% \times 96.2 + 3 \times 0.1}{96.2} = 0.060$$

$$\frac{\Delta E_6}{E_6} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 6}}{U_{kđ\ 6}} = 0.052 + \frac{0.5\% \times 84.7 + 3 \times 0.1}{84.7} = 0.061$$

$$\frac{\Delta E_7}{E_7} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 7}}{U_{kđ\ 7}} = 0.052 + \frac{0.5\% \times 74.9 + 3 \times 0.1}{74.9} = 0.061$$

$$\frac{\Delta E_8}{E_8} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 8}}{U_{kđ\ 8}} = 0.052 + \frac{0.5\% \times 67.7 + 3 \times 0.1}{67.7} = 0.061$$

$$\frac{\Delta E_9}{E_9} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 9}}{U_{kđ\ 9}} = 0.052 + \frac{0.5\% \times 58.4 + 3 \times 0.1}{58.4} = 0.062$$

$$\frac{\Delta E_{10}}{E_{10}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ\ 10}}{U_{kđ\ 10}} = 0.052 + \frac{0.5\% \times 48.1 + 3 \times 0.1}{48.1} = 0.063$$

$$\frac{\Delta E_{11}}{E_{11}} = \frac{\Delta k}{k} + \frac{\Delta U_{kđ1\ 1}}{U_{kđ\ 11}} = 0.052 + \frac{0.5\% \times 39.8 + 3 \times 0.1}{39.8} = 0.065$$

$\Delta E_i = \varepsilon_i E_i\ (mV)$

$\Delta E_{nđ1} = \varepsilon_1 . E_{nđ1} = 0.059 \times 5{,}09 = 0.300\ (mV)$

$\Delta E_{nđ2} = \varepsilon_2 . E_{nđ2} = 0.059 \times 4{,}41 = 0.260\ (mV)$

$\Delta E_{nđ3} = \varepsilon_3 . E_{nđ3} = 0.060 \times 3.98 = 0.239\ (mV)$

$\Delta E_{nđ4} = \varepsilon_4 . E_{nđ4} = 0.060 \times 3.51 = 0.211\ (mV)$

$\Delta E_{nđ5} = \varepsilon_5 . E_{nđ5} = 0.060 \times 3.21 = 0.193\ (mV)$

$\Delta E_{nđ6} = \varepsilon_6 . E_{nđ6} = 0.061 \times 2.82 = 0.170\ (mV)$

$\Delta E_{nđ7} = \varepsilon_7 . E_{nđ7} = 0.061 \times 2.50 = 0.153\ (mV)$

$\Delta E_{nđ8} = \varepsilon_8 . E_{nđ8} = 0.061 \times 2.26 = 0.138\ (mV)$

$\Delta E_{nđ9} = \varepsilon_9 . E_{nđ9} = 0.062 \times 1.95 = 0.121\ (mV)$

$\Delta E_{nđ10} = \varepsilon_{10} . E_{nđ10} = 0.063 \times 1.60 = 0.101\ (mV)$

$\Delta E_{nđ11} = \varepsilon_{11} . E_{nđ11} = 0.065 \times 1.33 = 0.086\ (mV)$

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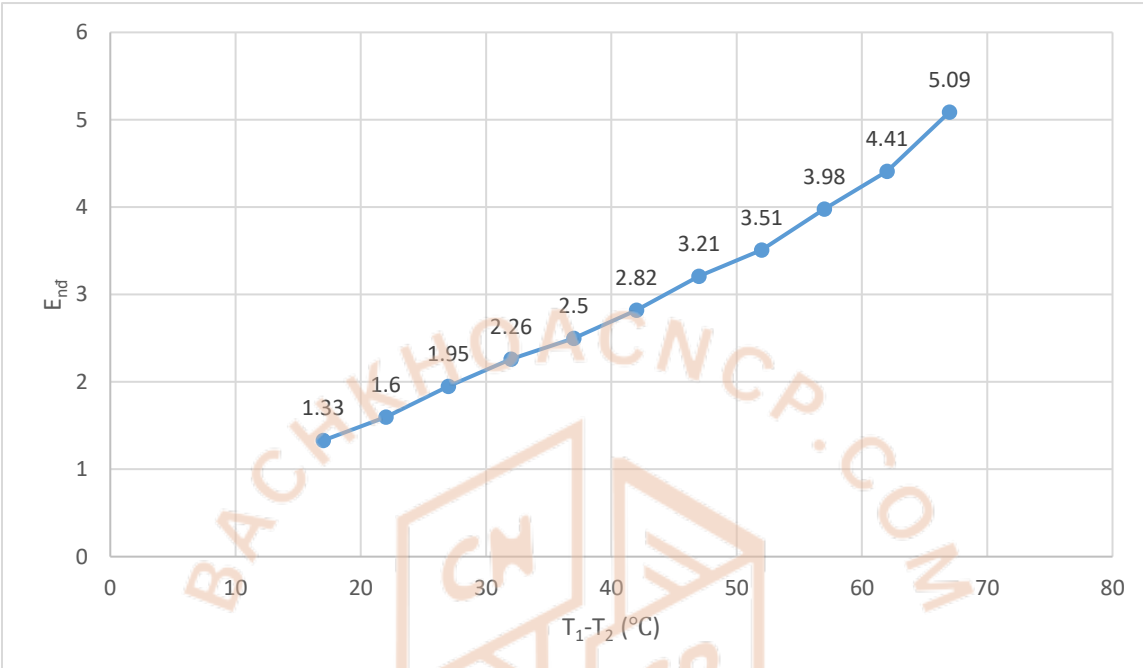
$\Delta(T_1 - T_2) = \Delta T_1 + \Delta T_2 = 1.0 + 0.3 = 1.3$ (°C)

Plot the graph of $E = C\ (T_1 - T_2)$ on a graph paper with scale: horizontal axis: 5°C/cm; vertical axis: 0.5mV/cm



Note:

- One point on the graph corresponds to one pair of E and $T = (T_1 - T_2)$ in Table 1. The error bars are represented by an error rectangle with its sides being $2\Delta T$ and $2\Delta E$.
- The graph must be a continuous and sharp line such that the centers of the error rectangles distribute equally on both sides of the line.

Determination of the thermocouple constant:

There are two methods to determine the thermocouple constant.

Method 1

In the working range of temperature, let the graph representing the function $E = C\ (T_1 - T_2)$ like the segment OA as shown in Figure Lab 4-4.

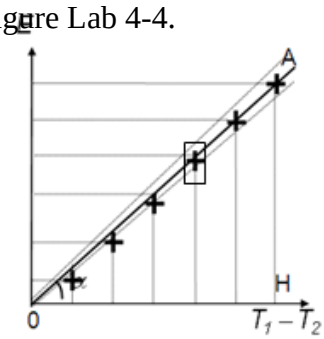


Figure Lab 4.4: Example graph

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Then, the average $\underline{C}$ and the absolute error ΔC of constant C are calculated as follows:

$$\overline{C} = \frac{\sum_{i=1}^{n}(T_{1i} - T_2)E_i}{\sum_{i=1}^{n}(T_{1i} - T_2)^2} = \frac{1568.42}{22154} = 0.071\left(\frac{mV}{°C}\right)$$

$\xi_{nđ1} = \bar{C}\,(T_{11} - T_2) = 0.071 \times 67 = 4.757$
$\xi_{nđ2} = \bar{C}\,(T_{12} - T_2) = 0.071 \times 62 = 4.402$
$\xi_{nđ3} = \bar{C}\,(T_{13} - T_2) = 0.071 \times 57 = 4.047$
$\xi_{nđ4} = \bar{C}\,(T_{14} - T_2) = 0.071 \times 52 = 3.692$
$\xi_{nđ5} = \bar{C}\,(T_{15} - T_2) = 0.071 \times 47 = 3.337$
$\xi_{nđ6} = \bar{C}\,(T_{16} - T_2) = 0.071 \times 42 = 2.982$
$\xi_{nđ7} = \bar{C}\,(T_{17} - T_2) = 0.071 \times 37 = 2.627$
$\xi_{nđ8} = \bar{C}\,(T_{18} - T_2) = 0.071 \times 32 = 2.272$
$\xi_{nđ9} = \bar{C}\,(T_{19} - T_2) = 0.071 \times 27 = 1.917$
$\xi_{nđ10} = \bar{C}\,(T_{110} - T_2) = 0.071 \times 22 = 1.562$
$\xi_{nđ11} = \bar{C}\,(T_{111} - T_2) = 0.071 \times 17 = 1.207$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(E_{nđi} - \xi_{nđi})^2 = \frac{1}{10} \times 0.225 = 0.023$$

$$(\Delta C)^2 = \frac{S^2}{\sum_{i=1}^{n}(T_{1i} - T_2)^2} = \frac{0.023}{22154} = 1.038\text{x}10^{-6}$$

$=> \Delta C = 0.0010$

Result:

$$\boxed{C = \overline{C} \pm \Delta C = 0.071 \pm 0.0010\left(\frac{mV}{°C}\right)}$$

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Method 2

Calculation of C

Let α be the tilt angle of segment *OA* as in Figure Lab 4-4 with respect to the horizontal axis, we have:

$$\overline{C} = tg\alpha = \frac{K_y \cdot AH}{K_x \cdot AH} = \frac{\frac{152.6}{30}}{67} = 0.076\left(\frac{mV}{°C}\right)$$

Calculation of error ΔC

Applying differential, we obtain:

$$dC = d(tg\alpha) = \frac{d\alpha}{cos^2\alpha} = (1 + tg^2\alpha)d\alpha$$

Since $C^2 = tg^2\alpha \ll 1$, we have:

$$\Delta C = (1 + C^2)\Delta\alpha = \Delta\alpha$$

Here, we can consider $\Delta\alpha$ as the angle between two lines starting from origin O and going through the two endpoints of the error segment $2\Delta E$. $\Delta\alpha$ is approximated as denoted as:

$$\Delta C \approx \Delta\alpha = \frac{2\Delta E}{OA}$$

$$OA = \sqrt{\overline{AH}^2 + \overline{OH}^2} = \sqrt{\left(\frac{152.6}{30}\right)^2 + 67^2} = 67.193$$

$$2\Delta E = 2 \times 0.300 = 0.600$$

Thus, according to equation 4, we obtain:

$$\Delta C \approx \Delta\alpha = \frac{2\Delta E}{OA} = \frac{0.600}{67.193} = 0.009$$

Result:

$$\boxed{C = \overline{C} \pm \Delta C = 0.076 \pm 0.009 \left(\frac{mV}{°C}\right)}$$

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EXTRA LAB 4:

CASSY LAB GUIDE

Figure Lab 4.5: Experimental setup

Figure Lab 4.5 shows the experimental setup using CASSY LAB system. The procedure for determining the thermocouple constant is as following steps:

1. Step 1: Click icon CASSY LAB on the desktop→ choose **Setting** board → select CASSY tab → click on UA1

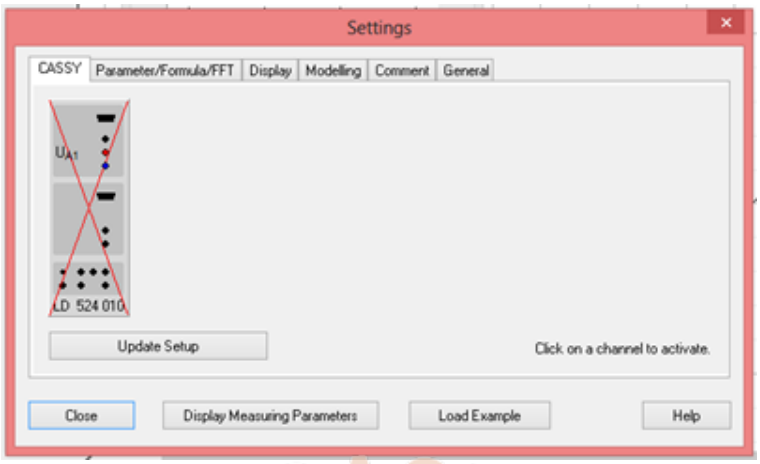

Figure Lab 4.6: Settings menu

Note: *After connecting CASSY LAB to the computer, this red cross will disappear.*

2. Step 2: "Sensor input settings" board: Select the parameters as

- Quantity: Choose Voltage UA1
- Zero point: Choose Left
- Meas. Range: 0 – 0.3V
- Record Measured Value: Choose Instantaneous Values

 Close

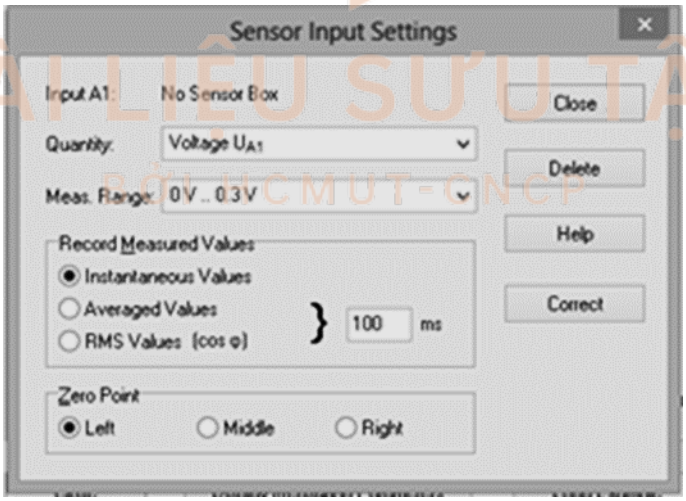

Figure Lab 4.7: Sensor input settings menu

3. Step 3: "Measuring Parameters" board: Select the parameters as

- Choose Manual Recording
- Close

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Figure Lab 4-8: Measuring parameter setting

4. Step 4: "Setting" board: Select the parameters as

- Choose "Parameter/Formula/FFT" tab Select Quantity

 Choose New Quantity

 Choose "Formula (time, date, n, t, U_{A1}, old="

 Enter the formula: 35+5*(n-1) if T_2 = 35℃

 Symbol: Enter T

 Unit: Enter ^oC

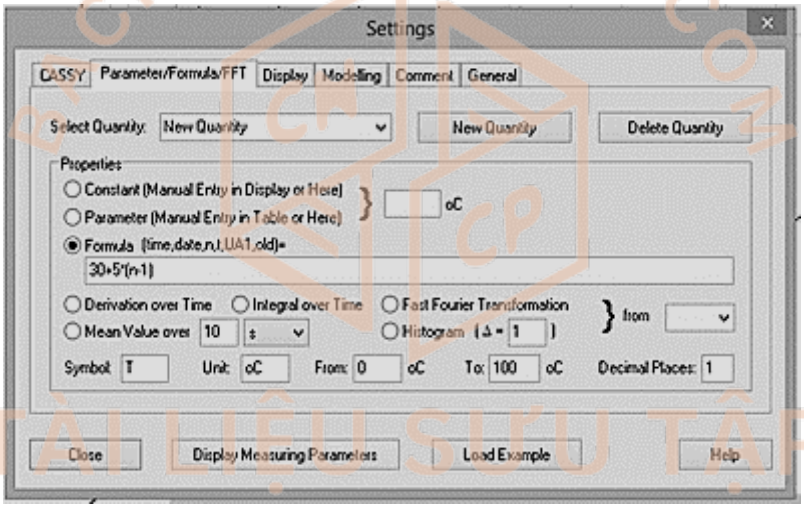

Figure Lab 4.9: Settings menu / Parameters tab

5. Step 5: "Display" tab, select Display: Display

- X-axis: Choose T
- Y-axis: Choose U_{A1}

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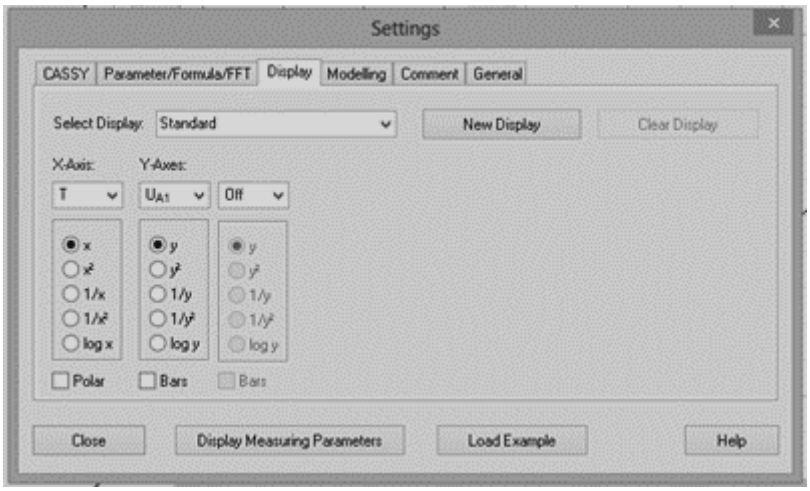

Figure Lab 4.10: Settings menu / Display tab

6. Step 6: Click on the main screen of CASSY LAB to draw the graph. Turn on the kettle of the experiment. At 35℃ press F9 on the keyboard, and press F9 while the thermometer shows an increment of 5℃until the temperature reach 95℃. Turn off the kettle.

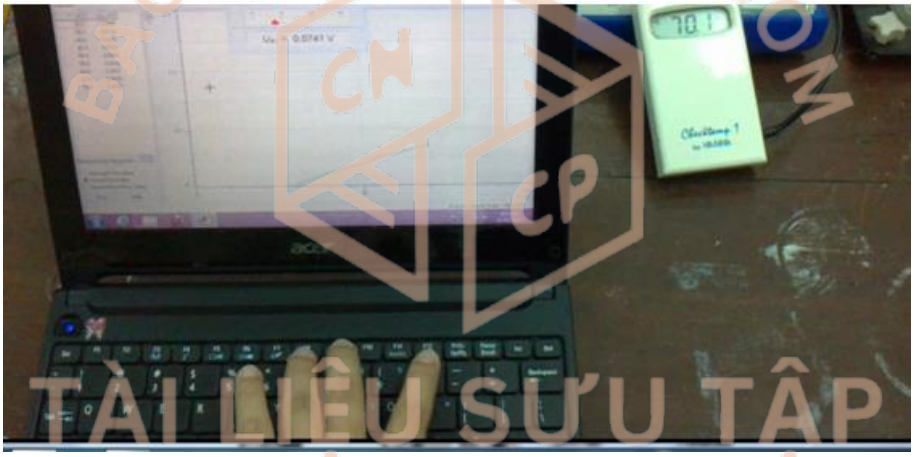

Figure Lab 4.11: Press F9 on the keyboard when the thermometer shows an increment of 5℃

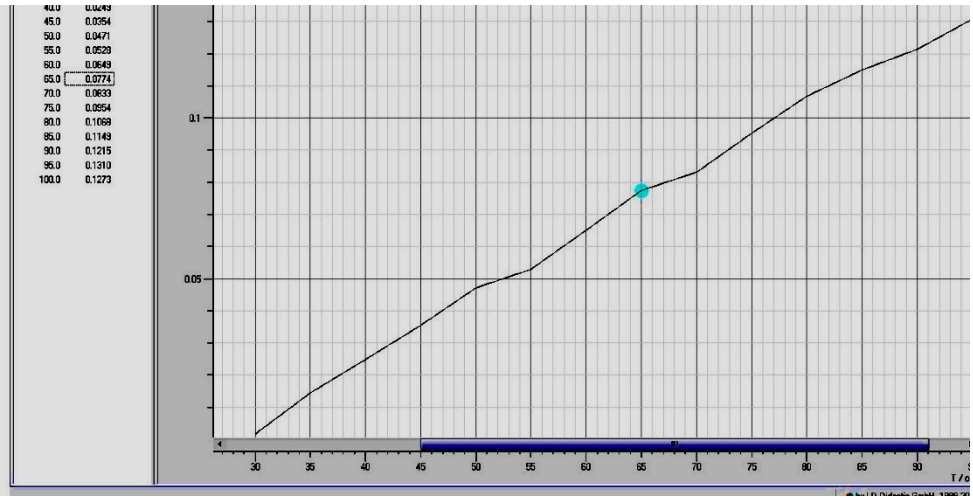

Figure Lab 4.12: Graph displays measurement results from 35℃ to 95℃

7. Step 7: On the main screen of CASSY LAB, right-click

• Choose Fit Function Choose Best-fit straight line

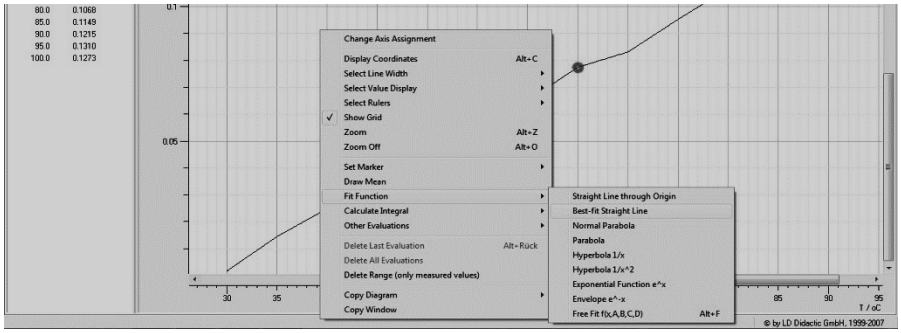

Figure Lab 4.13: Best fit straight line menu

Left click, click and drag from the beginning to the end of the graph, then we get the best line.

8. Step 8: Record the slope A and intercept B

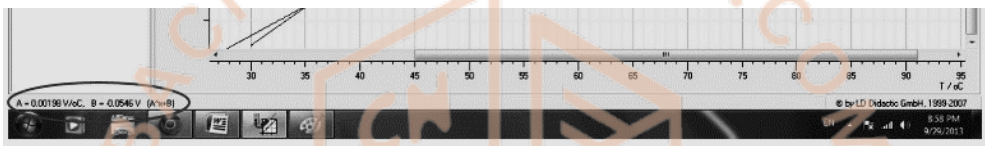

Figure 4.14: Record the slope A and intercept B

Get the data A, B (Figure 17) that is shown on the left corner of the screen Cassy LAB. Depend on the experiment step 3.1. We calculate the constant thermocouple.

The results: (C = A/k)

- **Method 1:**

$$C = \overline{C} \pm \Delta C = 0.071 \pm 0.001 \left(\frac{mV}{°C}\right)$$

- **Method 2:**

$$C = \overline{C} \pm \Delta C = 0.076 \pm 0.009 \left(\frac{mV}{°C}\right)$$

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